

LAB 03:DATA MANIPULATION

Objective:

Creating a student table and modifying records using UPDATE, and querying stored data using SELECT.

Introduction:

Structured Query Language (SQL) commands used to work with databases are broadly divided into two categories: Data Definition Language (DDL) and Data Manipulation Language (DML).

Data Definition Language (DDL):

DDL statements are used to define, alter, and manage the structure of database objects such as tables, indexes, and schemas. They do not manipulate the actual data stored within the tables. Common DDL commands include:

- CREATE: Builds a new table or database object.
- ALTER: Modifies the structure of an existing table.
- DROP:Deletes a table or database object permanently.

Data Manipulation Language (DML):

DML statements are used to manage and manipulate the actual data stored within database tables. They allow users to insert, modify, retrieve, and delete rows without altering the table's structural definition. Common DML commands include:

- INSERT — Adds new records to a table.
- UPDATE — Modifies existing data within a table.
- DELETE — Removes specific records from a table.
- SELECT — Retrieves and filters data from a table.

LAB TASK:

SELECT Queries

- Select * from stud table
- Select name from stud_table
- Select id from stud_table
- Select id, Name from stud_table where id=2
- Select name from stud_table where id>1
- SELECT name FROM stud_table WHERE name = 'Ali';
- Select name from stud_table where id<3

Procedure:

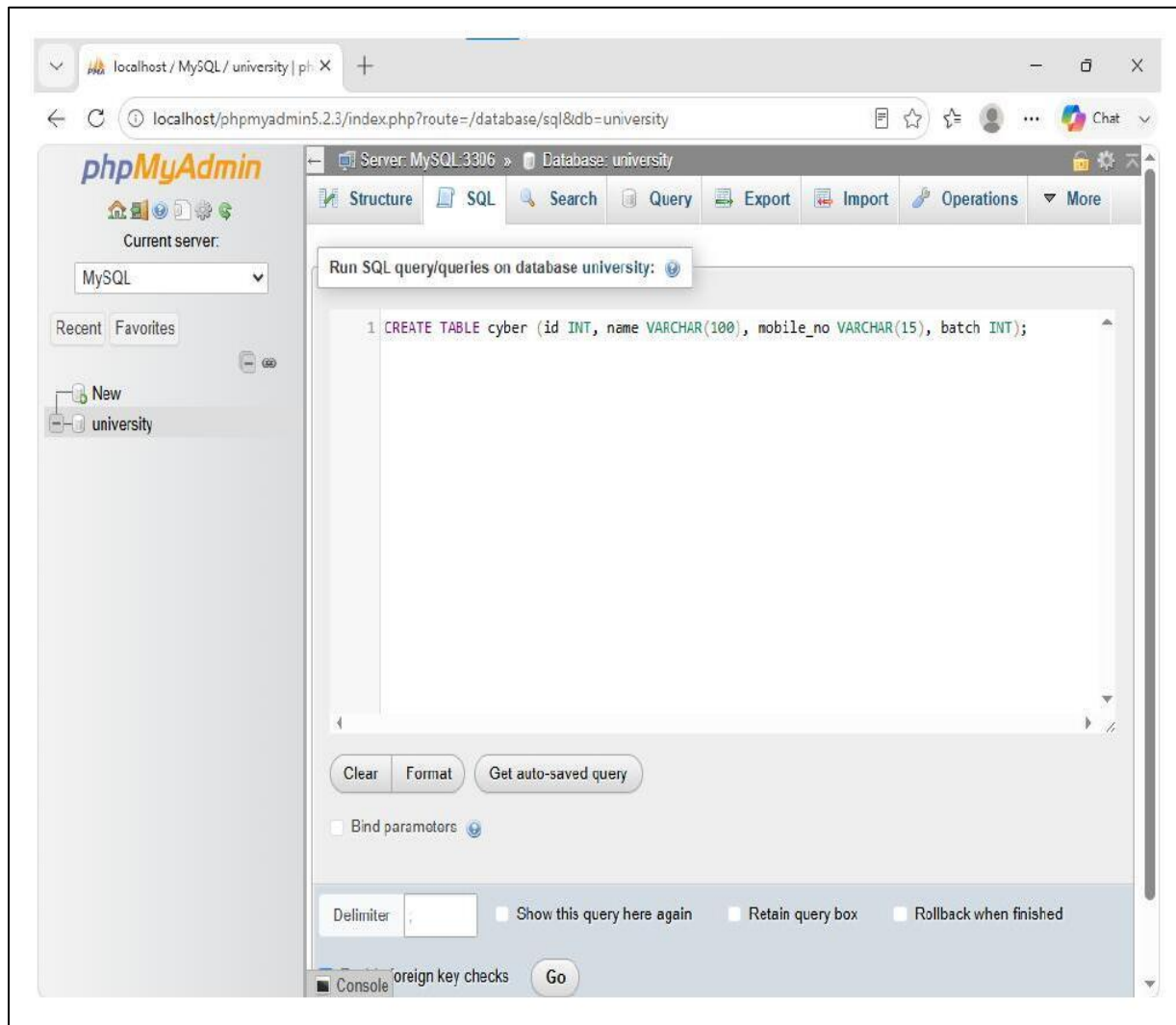
1. Start WampServer and open phpMyAdmin in the browser (localhost/phpmyadmin).
2. Select the "university" database.
3. Create the "cyber" table with the columns id, name, mobile_no and batch.
4. Insert four records into the "cyber" table using the INSERT INTO statement.
5. Modify the names of the existing records using the UPDATE command and display the table.
6. Run the SELECT queries one by one in the SQL tab.

Lab Task

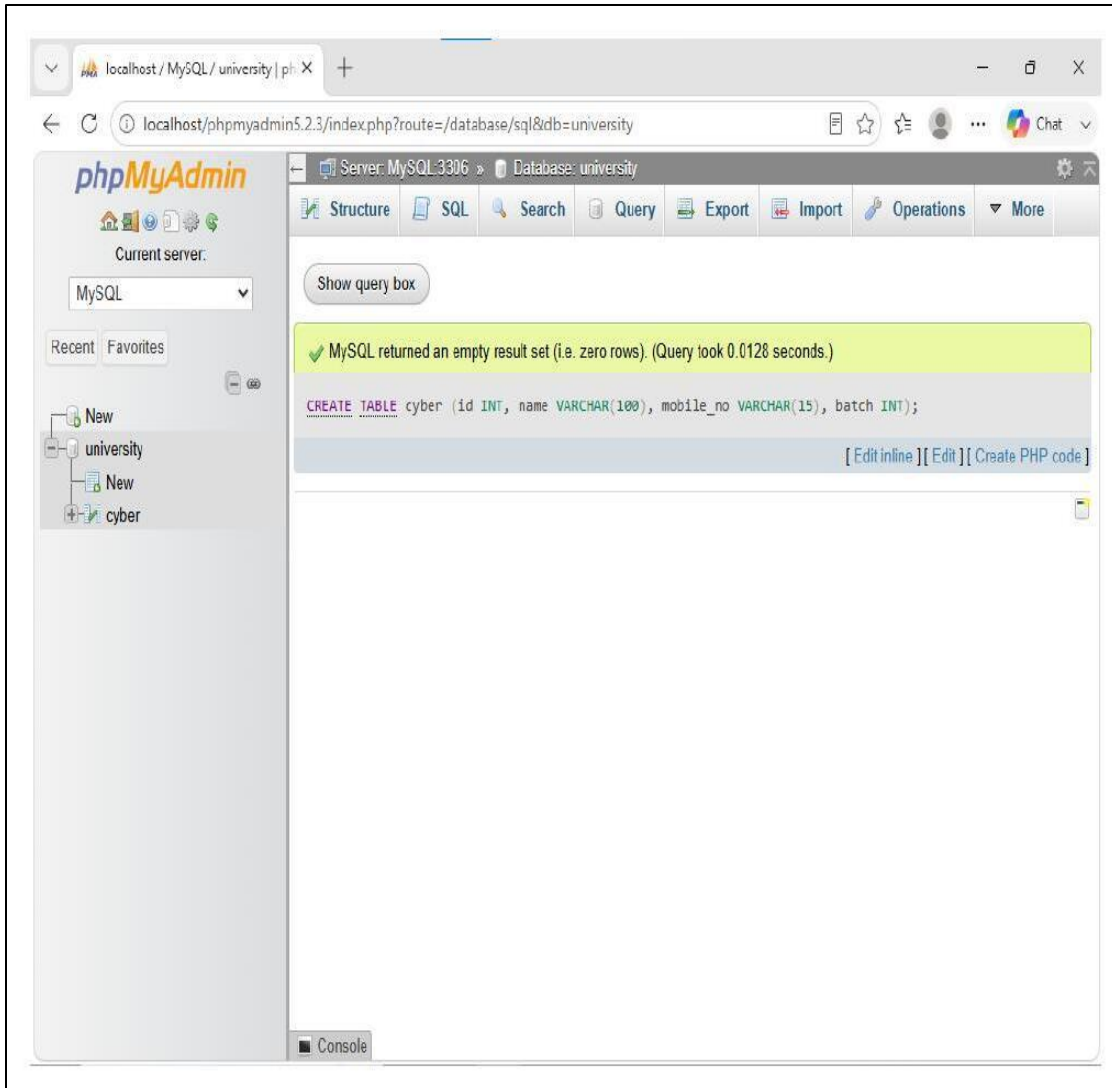
Step 1: Create Database Table

A table named "cyber" was created inside the "university" database using the following query, with four columns: id, name, mobile_no, and batch.

```
CREATE TABLE cyber (id INT, name VARCHAR(100), mobile_no VARCHAR(15), batch INT);
```

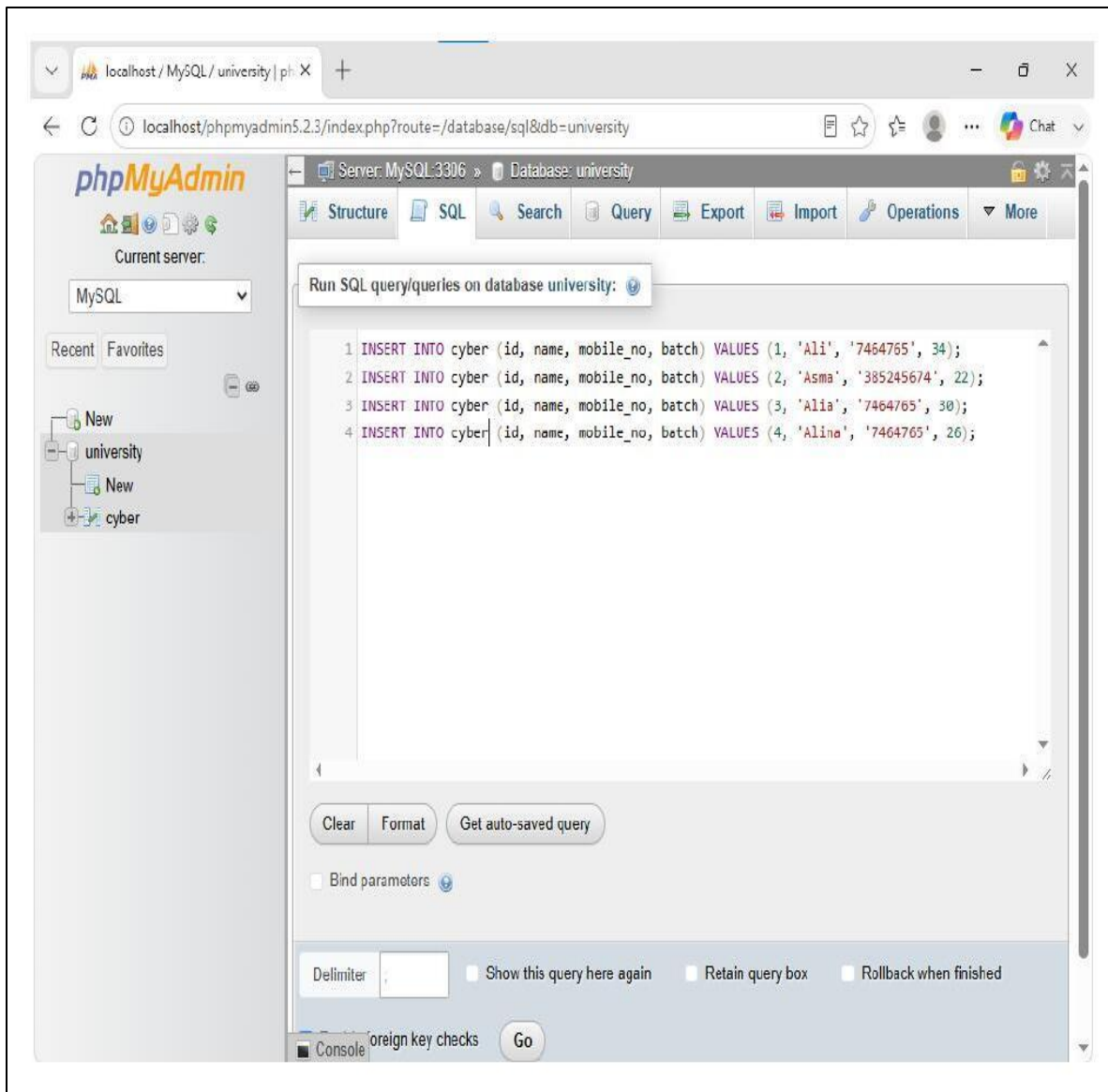


Output: Table created successfully



Step 2: Insert Records

Four records were added into the "cyber" table using INSERT INTO statements.



Output: Records inserted and displayed in the table

The screenshot shows the phpMyAdmin web interface. The browser address bar indicates the URL: `localhost/phpmyadmin5.2.3/index.php?route=/sql&pos=0&db=university&table=cyber`. The interface is divided into several sections:

- Left Sidebar:** Contains navigation links for 'New', 'university', 'New', and 'cyber'. It also shows 'Recent' and 'Favorites' sections.
- Top Toolbar:** Includes buttons for 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', and 'Privileges'.
- Message Bar:** A yellow bar at the top of the main content area states: 'Showing rows 0 - 3 (4 total, Query took 0.0011 seconds.)'.
- SQL Query:** The query entered is `SELECT * FROM `cyber``.
- Table Data:** The table 'cyber' is displayed with the following data:

id	name	mobile_no	batch
1	Ali	7464765	34
2	Asma	385245674	22
3	Alia	7464765	30
4	Alina	7464765	26

Below the table, there are controls for 'Show all', 'Number of rows' (set to 25), and a 'Filter rows' search box. At the bottom, there is a 'Query results operations' section with buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'. A 'Console' tab is visible at the very bottom.

Step 3: Update Records

The UPDATE command was used to modify the existing names in the "cyber" table.

The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL: `localhost/phpmyadmin5.2.3/index.php?route=/table/sql&db=university&table=cyber`. The interface shows the 'university' database selected, with the 'cyber' table chosen. The SQL tab is active, displaying four UPDATE queries:

```
1 UPDATE cyber SET name='Sara' WHERE id=1;
2 UPDATE cyber SET name='Asma' WHERE id=2;
3 UPDATE cyber SET name='Alia' WHERE id=3;
4 UPDATE cyber SET name='Asiya' WHERE id=4;
```

Below the queries, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. There is also a 'Get auto-saved query' button and a checkbox for 'Bind parameters'. At the bottom, there are options for 'Delimiter' (set to semicolon), 'Show this query here again', 'Retain query box', and 'Rollback when finished'. A 'Console' tab is visible at the bottom left.

Output: Table showing updated names

The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL: `localhost/phpmyadmin5.2.3/index.php?route=/sql&pos=0&db=university&table=cyber`. The interface shows the 'university' database selected, with the 'cyber' table chosen. A message states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this, a green bar indicates 'Showing rows 0 - 3 (4 total, Query took 0.0003 seconds.)'. The SQL query `SELECT * FROM `cyber`` is displayed. Below the query, there are options for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A 'Show all' checkbox is present, along with a 'Number of rows' dropdown set to 25 and a 'Filter rows' search box. The 'Extra options' section is expanded, showing a table with 4 rows and 4 columns: 'id', 'name', 'mobile_no', and 'batch'. The table data is as follows:

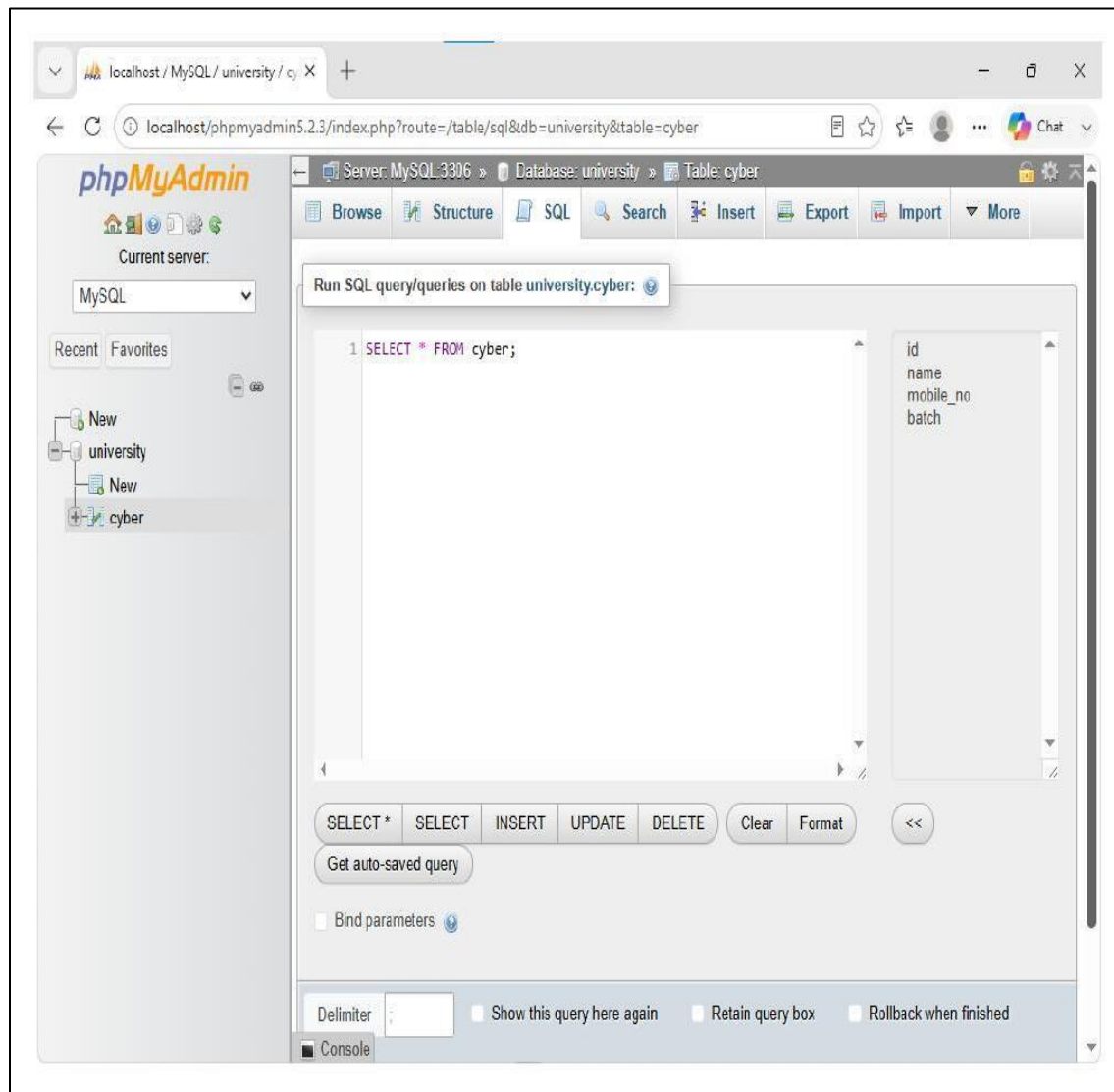
id	name	mobile_no	batch
1	Sara	7464765	34
2	Asma	385245674	22
3	Alia	7464765	30
4	Asiya	7464765	26

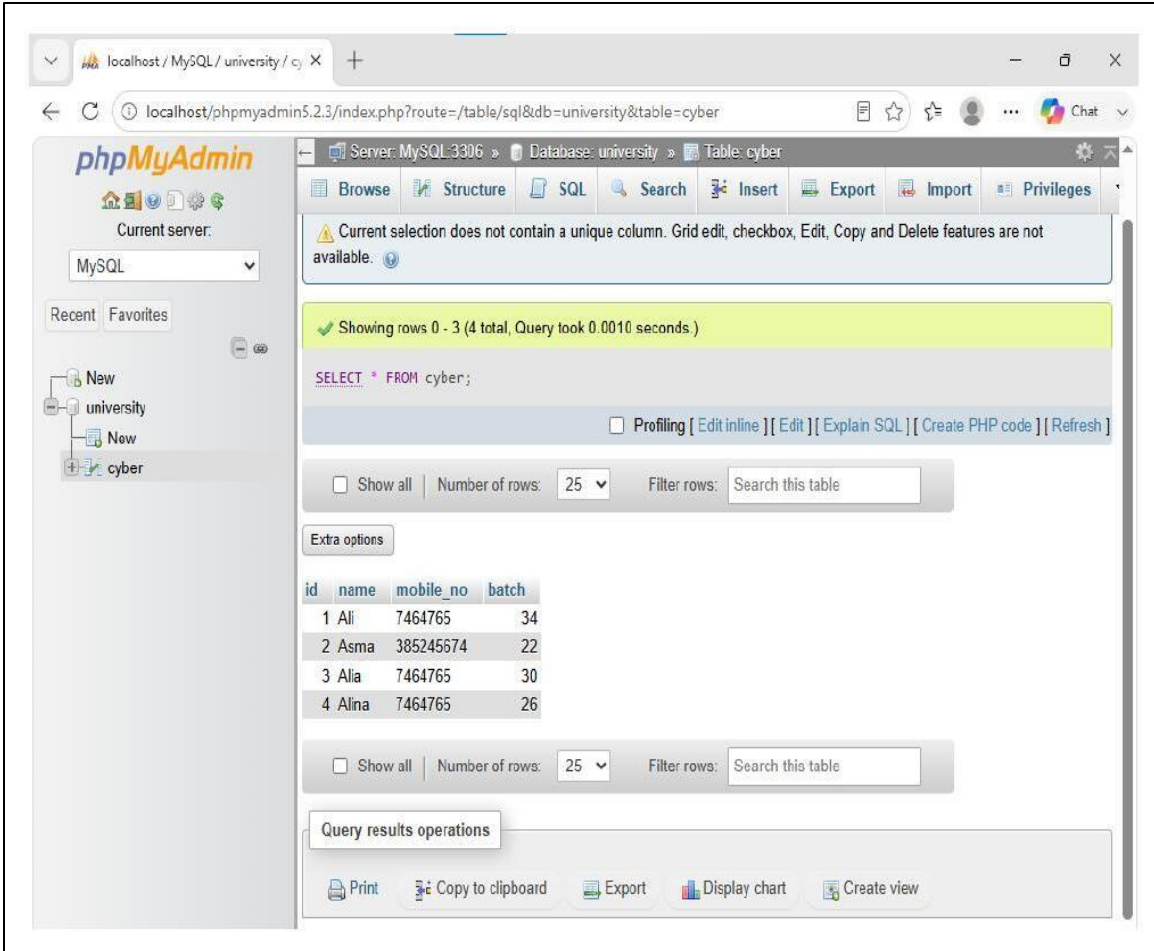
Below the table, there are again 'Show all', 'Number of rows' (25), and 'Filter rows' options. The 'Query results operations' section is expanded, showing buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'. At the bottom, there is a 'Console' tab.

Step 4: SELECT Queries

The following SELECT queries were run to retrieve and filter the stored data.

Query 1: *SELECT * FROM cyber;*



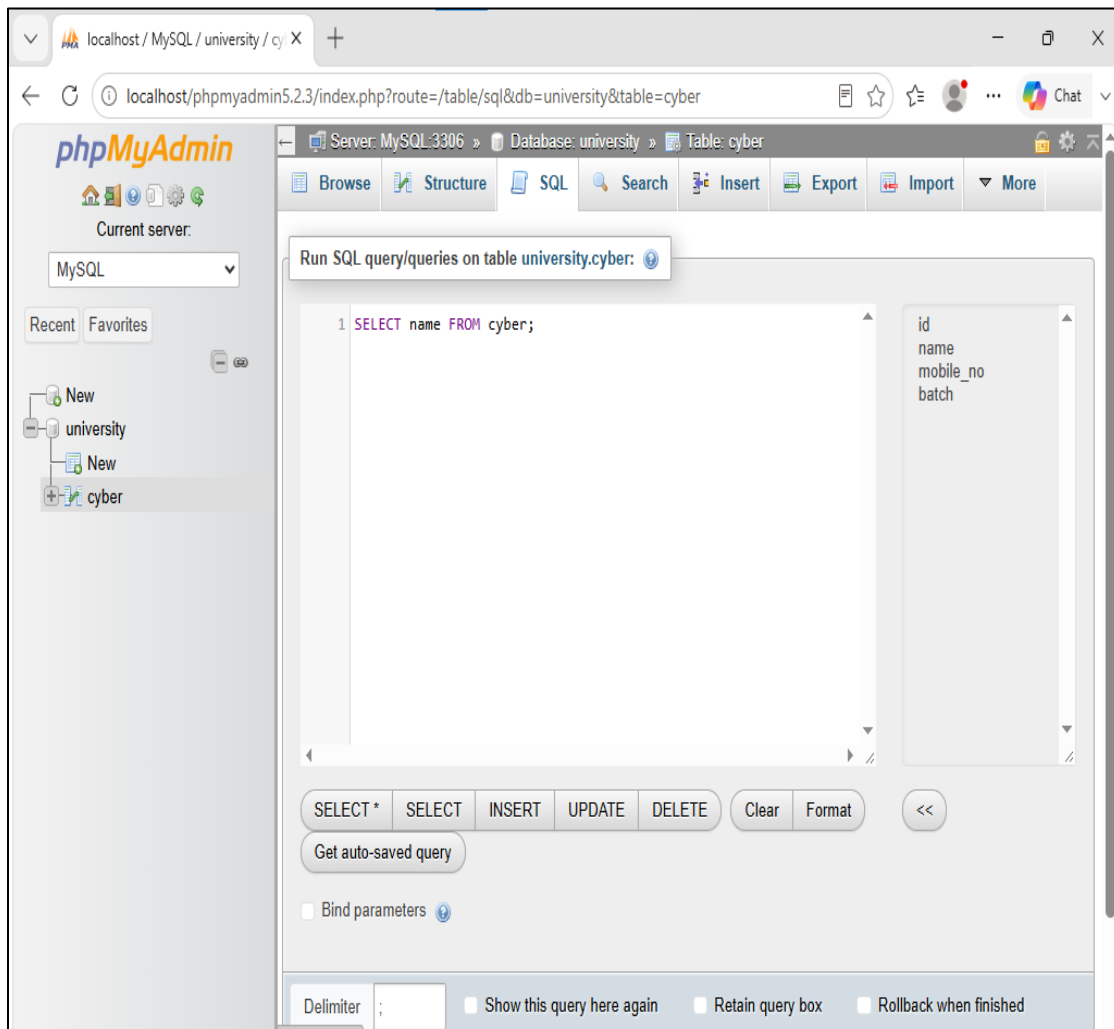


The screenshot displays the phpMyAdmin web interface. The browser address bar shows the URL: `localhost/phpmyadmin5.2.3/index.php?route=/table/sql&db=university&table=cyber`. The interface is divided into several sections:

- Left Sidebar:** Contains the 'Current server' dropdown set to 'MySQL', 'Recent' and 'Favorites' buttons, and a tree view showing the database structure: 'New' > 'university' > 'New' > 'cyber'.
- Top Navigation Bar:** Includes tabs for 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', and 'Privileges'.
- Main Content Area:**
 - A warning message: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.'
 - A green status bar: 'Showing rows 0 - 3 (4 total, Query took 0.0010 seconds)'.
 - A query editor showing the SQL: `SELECT * FROM cyber;` with options for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'.
 - Display options: 'Show all', 'Number of rows: 25', and a 'Filter rows' search box.
 - An 'Extra options' section.
 - A table with 4 rows and 4 columns:

id	name	mobile_no	batch
1	Ali	7464765	34
2	Asma	385245674	22
3	Alia	7464765	30
4	Alina	7464765	26
 - Another set of display options: 'Show all', 'Number of rows: 25', and a 'Filter rows' search box.
 - A 'Query results operations' section with buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

Query 2: *SELECT name FROM cyber;*



The screenshot displays the phpMyAdmin web interface in a browser. The address bar shows the URL: `localhost/phpmyadmin5.2.3/index.php?route=/table/sql&db=university&table=cyber`. The interface is for a MySQL server (3306) connected to the 'university' database, viewing the 'cyber' table.

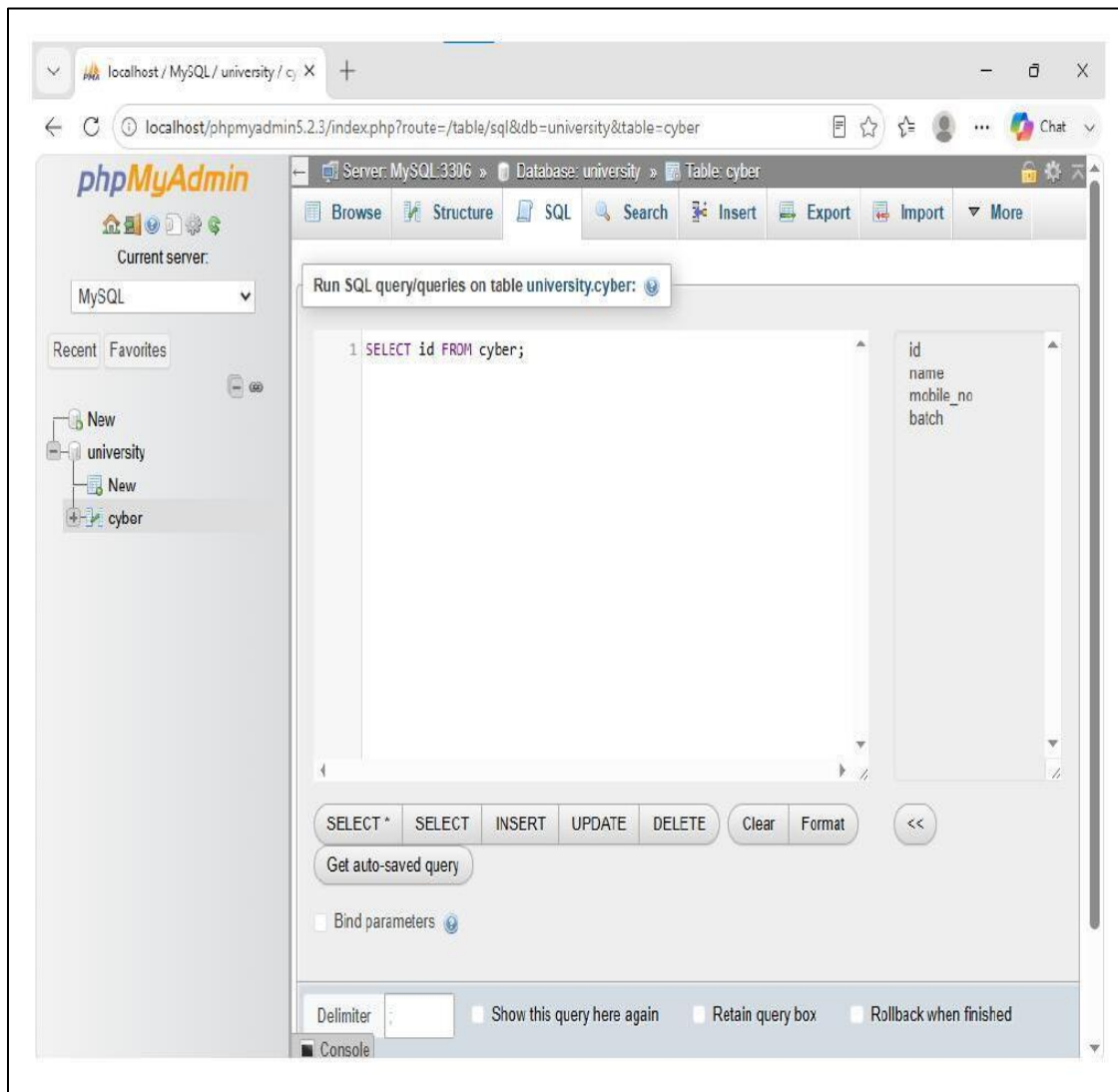
Left Sidebar: Shows the database structure with a tree view. The 'university' database is selected, and the 'cyber' table is highlighted under it. There are also links for 'New' and 'Recent'.

Main Content Area:

- Navigation Tabs:** Browse, Structure, SQL, Search, Insert, Export, Import, Privileges.
- Warning:** A yellow warning box states: "Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available."
- Query Results:** A green bar indicates "Showing rows 0 - 3 (4 total, Query took 0.0010 seconds.)". Below this, the SQL query `SELECT name FROM cyber;` is shown.
- Table Data:** A table with one column 'name' and four rows of data: Ali, Asma, Alia, and Alina.
- Query Results Operations:** A bar at the bottom of the table with buttons for Print, Copy to clipboard, Export, Display chart, and Create view.

Bottom Console: A tab labeled 'Console' is visible at the bottom of the interface.

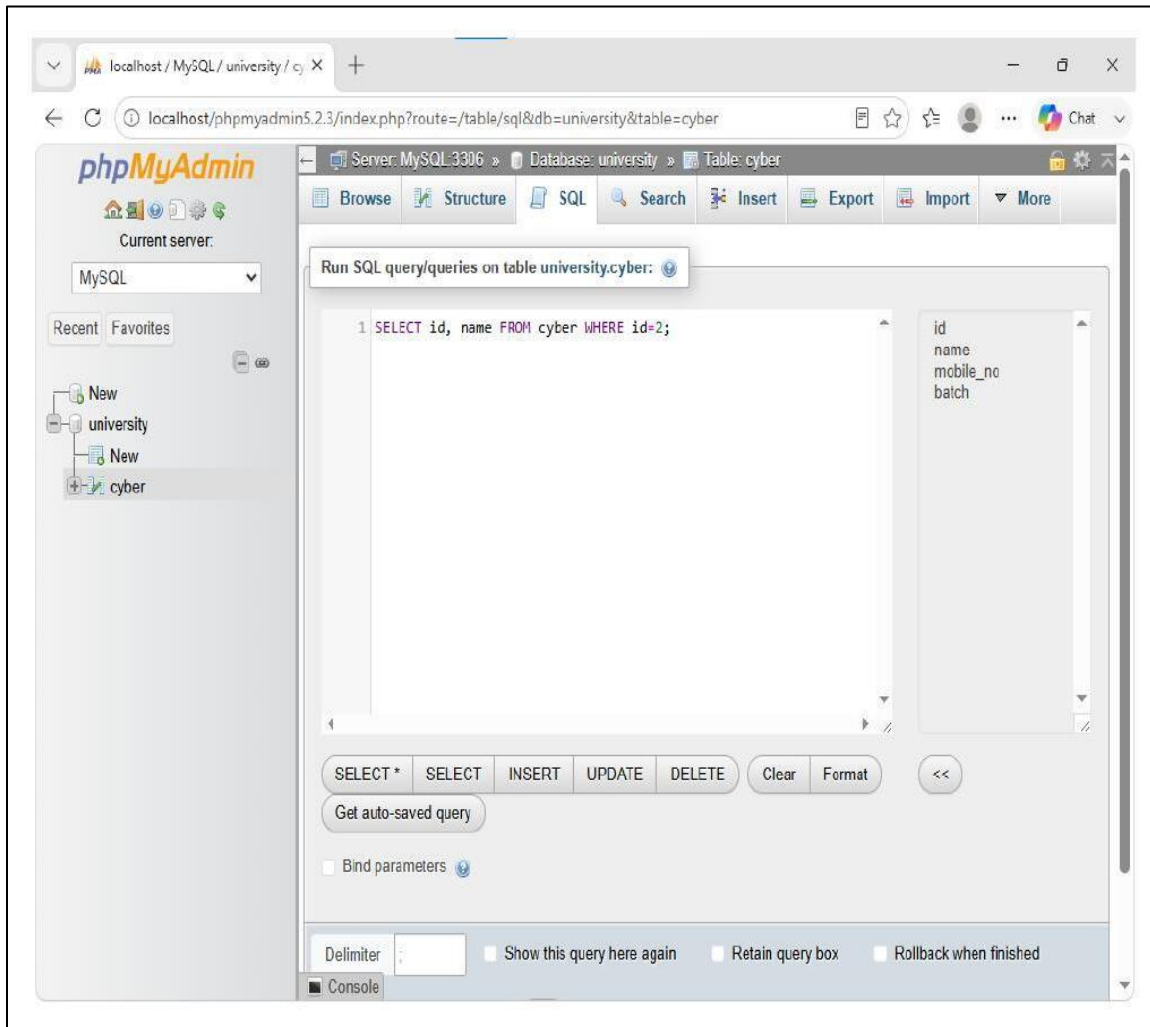
Query 3: *SELECT id FROM cyber;*



The screenshot displays the phpMyAdmin web interface in a Google Chrome browser. The address bar shows the URL: `localhost/phpmyadmin5.2.3/index.php?route=/table/sql&db=university&table=cyber`. The interface is divided into several sections:

- Left Sidebar:** Contains the phpMyAdmin logo, a 'Current server' dropdown set to 'MySQL', and a tree view showing the database structure: 'New' > 'university' > 'New' > 'cyber'.
- Top Toolbar:** Includes buttons for 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', and 'Privileges'.
- Message Bar:** A yellow warning message states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.'
- Query Results:** A green bar indicates 'Showing rows 0 - 3 (4 total, Query took 0.0003 seconds.)'. Below it, the SQL query `SELECT id FROM cyber;` is shown. A 'Profiling' section with links like '[Edit inline]', '[Edit]', '[Explain SQL]', '[Create PHP code]', and '[Refresh]' is also present.
- Table Controls:** A section with a 'Show all' checkbox, a 'Number of rows' dropdown set to '25', and a 'Filter rows' search box.
- Table Data:** A table with one column 'id' and four rows containing the values 1, 2, 3, and 4.
- Query Results Operations:** A section with buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.
- Bottom Console:** A 'Console' tab is visible at the bottom of the interface.

Query 4: *SELECT id, name FROM cyber WHERE id=2;*



The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL: `localhost/phpmyadmin5.2.3/index.php?route=/table/sql&db=university&table=cyber`. The interface is for the 'university' database, specifically the 'cyber' table. A message states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this, a green bar indicates 'Showing rows 0 - 0 (1 total, Query took 0.0011 seconds.)'. The SQL query executed is `SELECT id, name FROM cyber WHERE id=2;`. The results are displayed in a table with two columns: 'id' and 'name'. The first row shows '2' and 'Asma'. Below the table, there are options for 'Show all', 'Number of rows' (set to 25), and a 'Filter rows' search box. At the bottom, there are links for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'. The left sidebar shows the database structure with 'university' selected and 'cyber' highlighted.

localhost / MySQL / university / cyber

localhost/phpmyadmin5.2.3/index.php?route=/table/sql&db=university&table=cyber

Server: MySQL:3306 » Database: university » Table: cyber

Browse Structure SQL Search Insert Export Import Privileges

Show query box

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✓ Showing rows 0 - 0 (1 total, Query took 0.0011 seconds.)

`SELECT id, name FROM cyber WHERE id=2;`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

id	name
2	Asma

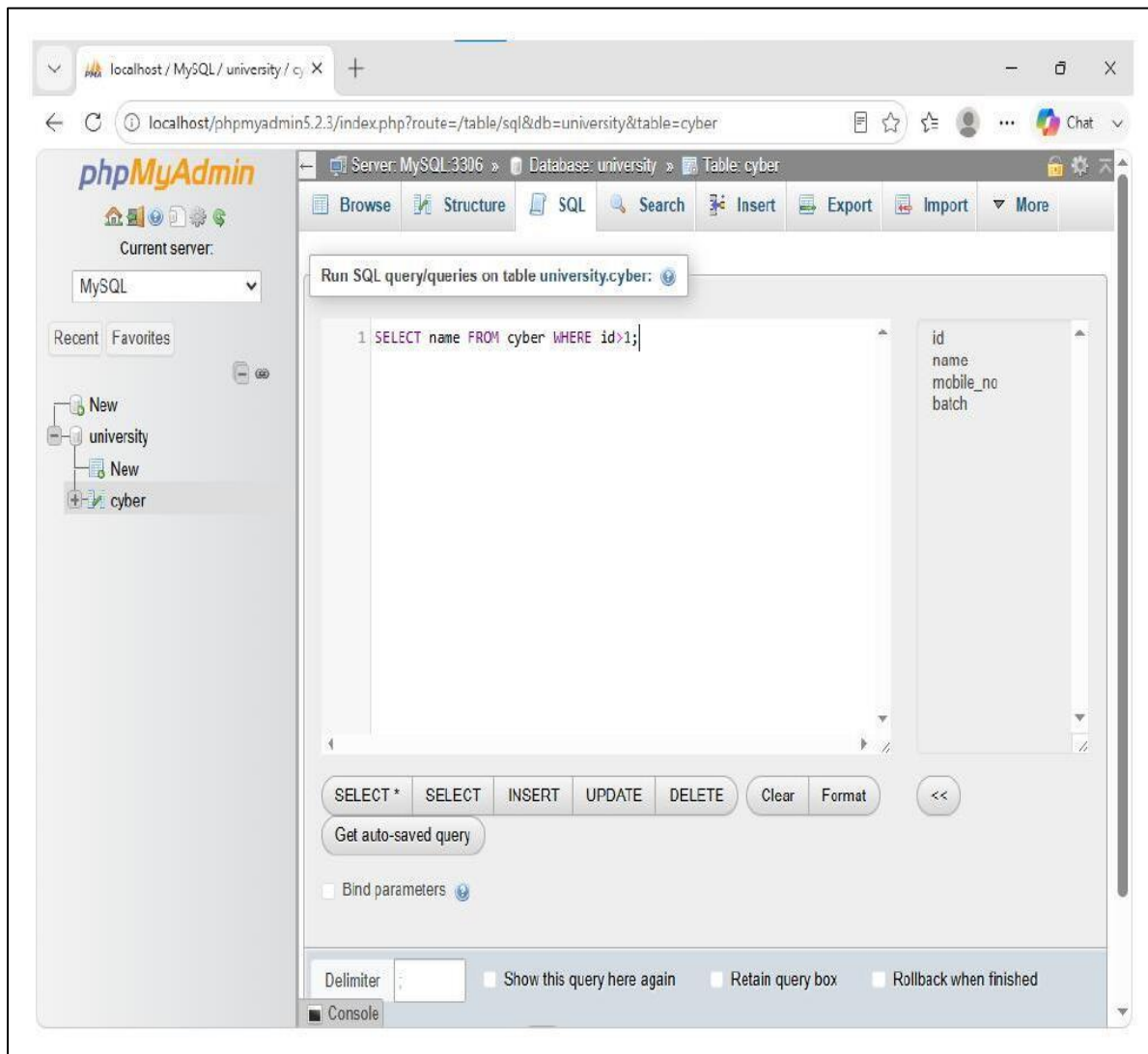
☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

Console

Query 5: *SELECT name FROM cyber WHERE id>1;*



The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL: `localhost/phpmyadmin5.2.3/index.php?route=/table/sql&db=university&table=cyber`. The interface is for a MySQL server running on localhost:3306, connected to the 'university' database, viewing the 'cyber' table.

A warning message at the top states: "Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available."

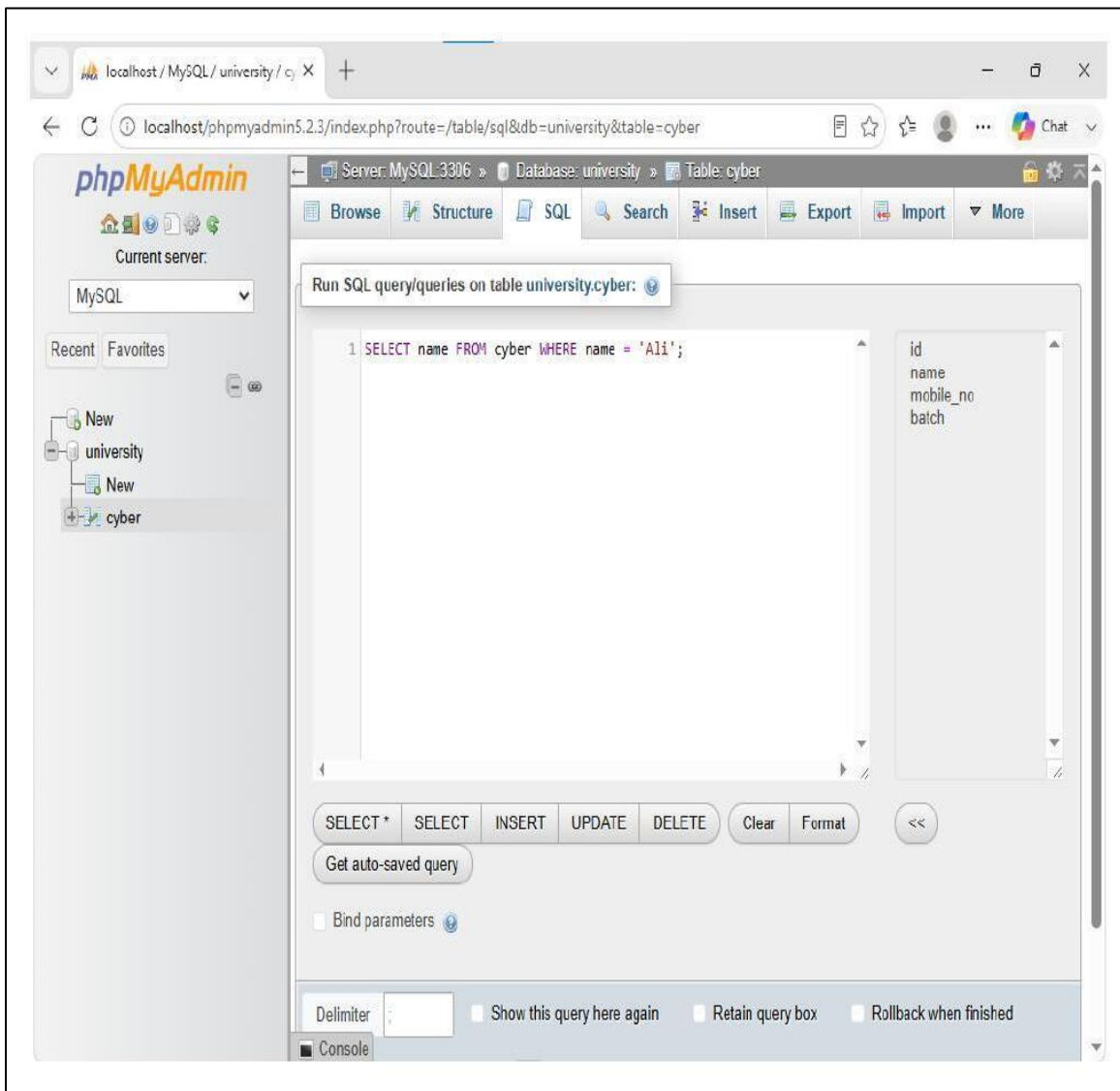
The main content area shows a successful query execution: "Showing rows 0 - 2 (3 total, Query took 0.0080 seconds.)". The SQL query entered is: `SELECT name FROM cyber WHERE id>1;`. Below the query, there are links for "Profiling", "Edit inline", "Edit", "Explain SQL", "Create PHP code", and "Refresh".

Below the query, there are controls for "Show all", "Number of rows" (set to 25), and a "Filter rows" search box.

The "Extra options" section shows the results of the query as a table with one column, "name", and three rows: "Asma", "Alia", and "Alina".

At the bottom, there is a "Query results operations" section with buttons for "Print", "Copy to clipboard", "Export", "Display chart", and "Create view".

Query 6: *SELECT name FROM cyber WHERE name = 'Ali';*



The screenshot displays the phpMyAdmin web interface in a browser. The address bar shows the URL: `localhost/phpmyadmin5.2.3/index.php?route=/table/sql&db=university&table=cyber`. The interface is for a MySQL server (localhost:3306) and the 'university' database. The selected table is 'cyber'. A message states: "Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available." The query executed is: `SELECT name FROM cyber WHERE name = 'Ali';`. The result shows 1 row: 'Ali'. The interface includes a left sidebar with a database tree, a top toolbar with options like Browse, Structure, SQL, Search, Insert, Export, Import, and Privileges, and a bottom section for query results operations (Print, Copy to clipboard, Export, Display chart, Create view).

Server: MySQL:3306 » Database: university » Table: cyber

Tools: Browse Structure SQL Search Insert Export Import Privileges

Show query box

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 0 (1 total, Query took 0.0010 seconds.)

```
SELECT name FROM cyber WHERE name = 'Ali';
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

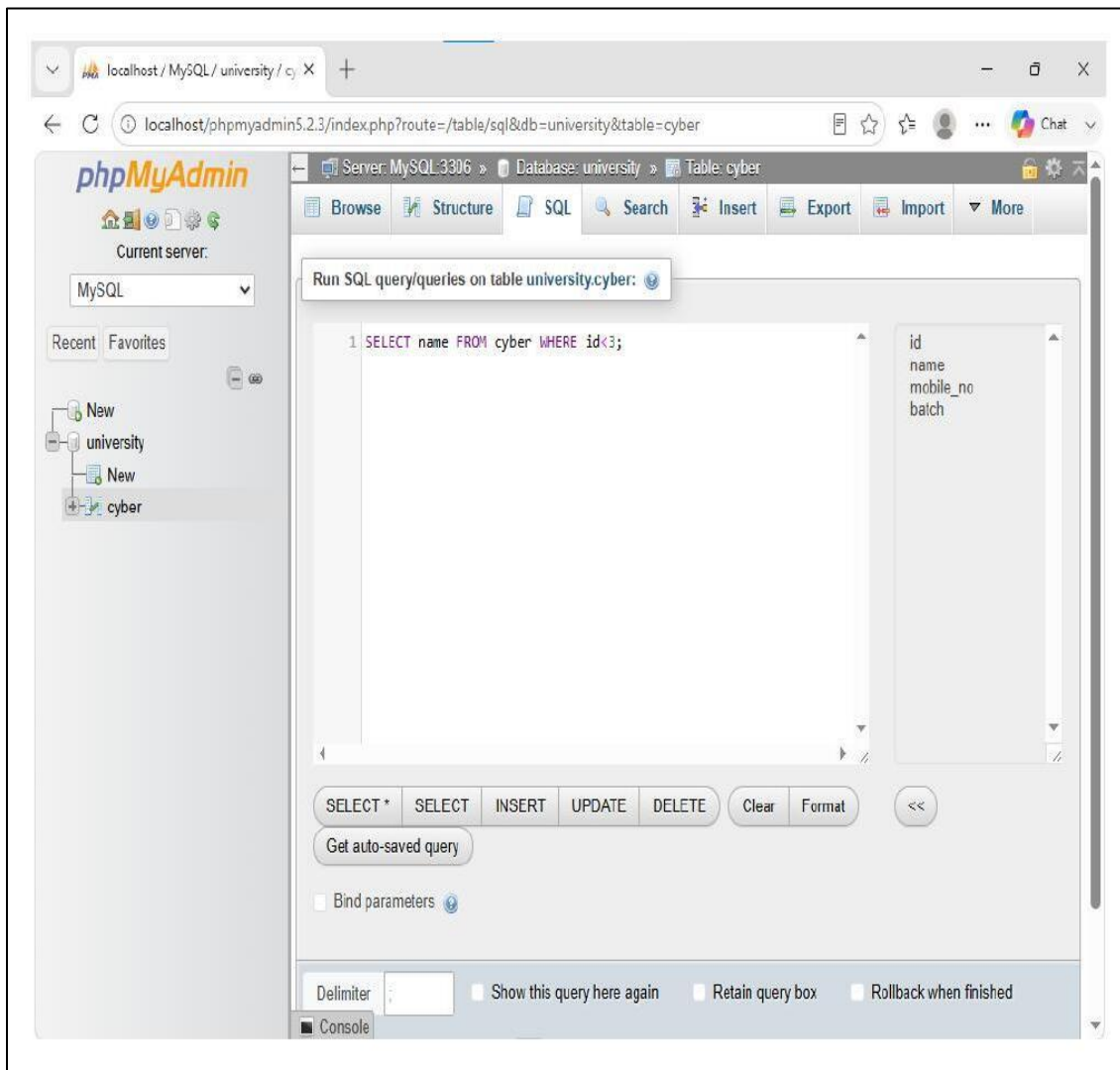
Extra options

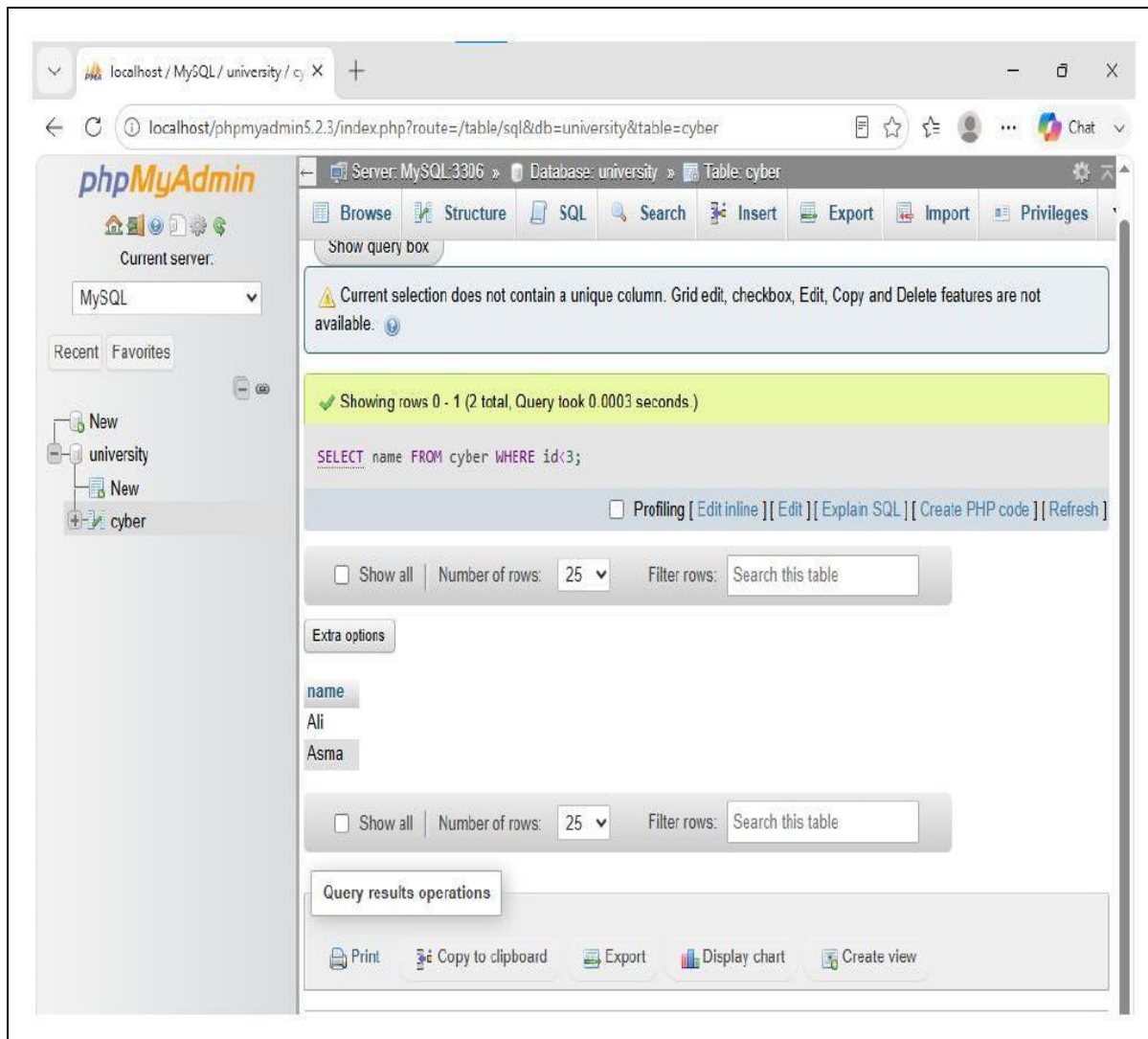
name
Ali

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Query 7: *SELECT name FROM cyber WHERE id<3;*





Results:

The "cyber" table was created successfully inside the "university" database with four columns: id, name, mobile_no and batch. Four records were inserted into the table, and the UPDATE command was then used to modify the names of the existing records, which was confirmed by displaying the table again.

The SELECT queries gave the following results:

- SELECT * FROM cyber returned all four records with all columns.
- SELECT name returned only the name column (4 rows).
- SELECT id returned only the id column (values 1 to 4).
- SELECT id, name WHERE id=2 returned one row: 2, Asma.
- SELECT name WHERE id>1 returned three rows: Asma, Alia and Alina.
- SELECT name WHERE name = 'Ali' returned one row: Ali.

- `SELECT name WHERE id<3` returned two rows: Ali and Asma.

Conclusion:

Database languages are divided based on their roles: Data Definition Language (DDL) focuses on building, modifying, and managing the overall structure of a database, while Data Manipulation Language (DML) handles the actual data stored inside those structures by allowing users to add, edit, delete, and retrieve records. This lab provided hands-on practice with `CREATE`, `INSERT`, `UPDATE`, and `SELECT` statements, and helped build a clearer understanding of how data can be manipulated and queried within a MySQL database using phpMyAdmin.